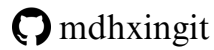


MIDHLAJ P.B.

STEM & Robotics Trainer | EduTech Enthusiast

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mdhxingit



Midhlaj P. Basheer



Portfolio

PROFESSIONAL SUMMARY

BCA graduate specializing in Embedded Systems, Robotics, and IoT with hands-on experience in building real-world robotic and sensor-based systems. Passionate about STEM education and simplifying complex technical concepts for young learners through interactive, activity-based learning. Experienced in mentoring students, conducting educational outreach sessions, and developing engaging robotics and AI demonstrations. Seeking to contribute technical expertise and teaching enthusiasm as a STEM & Robotics Trainer.

AREAS OF EXPERTISE

STEM Teaching & Mentoring: Age-appropriate instruction (Grades 4–8) | Student engagement

Robotics & AI Domains: Robotics design & Automation | Computer vision (OpenCV) | IoT sensor networks | AI/ML fundamentals

Embedded Platforms: ESP32 (ESP-IDF, FreeRTOS) | Raspberry Pi | Arduino

Programming: C / C++ | Python

Communication Protocols: UART | I2C | LoRa 433 MHz

Soft Skills: Public speaking | Presentation & demonstration | Communication & negotiation | Multilingual

Tools: Git | ESP-IDF toolchain | OpenCV | Logic analyser | VS Code

TEACHING & MENTORING EXPERIENCE

English Language & Student Mentor:

2025 - 2026 | Manglore

Literary Club, Yenepoya Institute of Arts, Science, Commerce & Management

- Conducted English vocabulary and grammar sessions for Grades 4–8 students.
- Used interactive teaching methods to improve student engagement and participation.
- Adapted teaching approaches based on student age groups and learning levels.
- Organized academic events and student engagement activities with the literary club.

Robotics & Ai Trainer

2026 | Palakkad

Industrial Robotics Institute

- Conducted Robotics, AI, Electronics, and IoT sessions for Grades 1-8 using interactive, hands-on, and age-appropriate teaching methods.
- Designed and delivered project-based STEM activities, guiding students through building, programming, testing, troubleshooting, and presenting technical projects.
- Mentored students in problem-solving, logical and computational thinking, creativity, and practical application through robotics and STEM activities.

PROFESSIONAL EXPERIENCE

Embedded Engineer Intern

01/2026 – 04/2026 | Infopark, Thrissur

Estro Tech Robotics & Innovations Pvt Ltd.

- Developed C firmware for ESP32-based IoT systems using I2C and UART communication.
- Debugged hardware-software integration issues using serial monitoring and logic analysis.
- Assisted in firmware optimization and prototype testing for embedded IoT devices.
- Worked on robotics and IoT hardware for real-world STEM demonstration systems.

ACADEMIC BACKGROUND

Yenepoya Institute of Arts, Science, Commerce & Management, Mangalore

2024 - 2026

Bachelor of Computer Applications (BCA) — AI/ML, Robotics & IoT Specialization

CGPA: 8.9

PROJECTS

Automated Object Sorting Robot [↗](#)

Developed a robotic system using Raspberry Pi, OpenCV, servo motors, and webcam-based color detection to sort objects by color with 90%+ accuracy. Implemented advanced automation logic for autonomous operation, showcasing strong skills in embedded systems and sensor integration.

Centralized Private Network using LoRa [↗](#)

Built a private IoT sensor network using ESP32 and custom raw LoRa protocol. Requires no Internet or SIM. Implemented FreeRTOS multi-task architecture with TDMA-based packet aggregation and CRC16 validation across 433 MHz ISM band. Main gateway hosts a WiFi AP serving a real-time WebSocket dashboard with online/offline node monitoring no internet dependency.

CERTIFICATIONS

- IoT Fundamentals & STEM Robotics — Infosys SpringBoard
- Python, AI/ML & Deep Learning Specializations — Coursera
- Microsoft Certified: Azure Fundamentals (AZ-900) — Microsoft